

Mohammad Sonji

Computer Science PhD Candidate

Research Interests

My research is broadly in computer systems, focusing on GPU architecture/kernel optimization, high-performance computing, and interconnect/memory systems.

Education

Aug 2024 - **Doctor of Philosophy, Computer Science, University of Virginia (UVA)**

May 2029 : CGPA: 3.9

Advisor: **Dr. Adwait Jog**

Jan 2022 - **Master of Science, Computer Science, American University of Beirut (AUB)**

Dec 2023 : CGPA: 3.78, Thesis title: Predicting If Executing Applications Are Near Completion

Advisor: **Dr. Izzat El Hajj**

2018 - 2021 : **Bachelor of Science, Computer Science, Beirut Arab University (BAU)**

CGPA: 3.76, Class rank: 3rd, Graduated with honors.

Experience

Research

Aug 2024 - **Research Assistant, UVA**

present : I joined **Dr. Adwait Jog's** Insight Computer Architecture Lab as a research assistant.

Jun 2022 - **Research Assistant, AUB & Hewlett Packard Labs, HPE**

Jul 2024 : I joined **Dr. Izzat El Hajj's** team in collaboration with Hewlett Packard Labs (HPE) led by **Dr. Dejan Mилојић** and his team.

Teaching

Teaching Assistant, UVA

Spring 26 : CS6501: GPU Architectures

Fall 25 : CS4444: Introduction to Parallel Computing (CUDA)

Teaching Assistant, AUB

Fall 23-24 : CMPS221: Computer Organization & Design

Spring 22-23 : CMPS202: Intermediate Programming with Data Structures

Fall 22-23 : CMPS224/CMPS396AA: GPU Computing (CUDA)

Spring 21-22 : CMPS200: Introduction to Programming

CMPS212: Intermediate Programming with Data Structures

Selected Academic Projects

2022 : **GPU Computing**

I designed a CUDA application in C/C++ for computing the Jaccard similarity among vertices in a graph, implementing four distinct versions of the code, each integrating one or more novel optimizations.

I was able to achieve the fastest execution time among my classmates.

2022 : **Compiler Construction**

Using the LLVM compiler and its Clang frontend, I implemented the following tasks in C++:

- Source-to-source compiler as a recursive AST visitor in Clang
- Code generator as a non-recursive AST visitor in Clang
- Aggressive Dead Code Elimination optimization.

Publications

UVA

ISPASS'26 **Hibiscus: End-to-end Architectural Simulation Framework for Hybrid SFQ/CMOS-Memory Compute Systems**, Ryan Marsala, Yerzhan Mustafa, Prabhath Tangella, Mohammad Sonji, George Michelogiannakis, Selcuk Kose, Adwait Jog and Mehmet E. Belviranli, In the Proceedings of IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), 2026

ISPASS'25 **Dissecting Performance Overheads of Confidential Computing on GPU-based Systems**, Yang Yang, Mohammad Sonji, Adwait Jog, In the Proceedings of IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), Ghent, Belgium, 2025

AUB

ICPE'26 **Are We There Yet? Predicting if Executing Applications are Near Completion**, M. Sonji, M. Baydoun, S. Diab, A. Nassereldine, P. Bruel, A. Dhakal, R. Enriquez, G. Rattihalli, D. Tootaghaj, G. Renaud, B. Chapman, F. Salem, E. Frachtenberg, D. Milojevic, I. Hajj, In the Proceedings of ACM/SPEC International Conference on Performance Engineering (ICPE), Florence, Italy, 2026

iWAPT'25 **Predicting Performance Variability**, M. Baydoun, M. Sonji, P. Bruel, D. Milojevic, E. Frachtenberg, I. El Hajj, In the Proceedings of IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), 2025

GrAPL'25 **Serverless Graph Analytics on Multi-Instance GPU**, M. Sonji, M. Baydoun, A. Dhakal, G. Rattihalli, D. Milojevic, I. El Hajj, In the Proceedings of IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), 2025

In progress

- **First-author work submitted to HPDC'26, UVA**, In this work, we identify and address a multi-tenancy architectural issue on Multi-instance GPU enabled GPUs that can substantially impact the performance of corunning interconnect-intensive applications.

Awards

2024 : **Fellowship**, UVA

2024 : **Henri Qais Naccache Best Thesis Award**, AUB

2018-2021 : **Scholarships**, BAU

Technical skills

Parallel, Distributed, & GPGPU Programming: CUDA, MPI, Pthreads

Programming Languages: C, C++, Python, Bash

Systems & Tools: Linux, Docker, Kubernetes, Knative, Git

NVIDIA GPU Partitioning: Multi-Instance GPU (MIG), Multi-Process Service (MPS)

LLM Inference, vLLM, HuggingFace